KEERTHANAN PS

Cybersecurity Researcher | Malware Analyst | Exploit Developer | Systems Engineer

keerthanan@tutanota.de | +91 8891975883 | Thiruvananthapuram, Kerala

Portfolio | GitHub | TryHackMe | LinkedIn | Research Blog

PROFESSIONAL SUMMARY

Cybersecurity researcher and systems developer with hands-on experience spanning malware analysis, reverse engineering, exploit development, operating system engineering, and security infrastructure. Experienced in analysing Windows PE and Linux ELF binaries, researching software vulnerabilities, and developing low-level system software using C and x86 Assembly.

Developing independent engineering projects including NanoSec OS, a custom operating system, and Petti, a Linux container runtime written in Go. Strong technical interests in kernel architecture, memory management, network protocols, containerisation, threat detection, and security engineering. Maintains an independent malware research blog documenting technical investigations and security research.

INDEPENDENT RESEARCH & DEVELOPMENT

Independent Cybersecurity Researcher & Developer *Sep 2020 – Present*

- Conduct independent research and development across cybersecurity, systems programming, vulnerability research, and security engineering.
- Develop custom security tools, automation utilities, and Python-based applications for reconnaissance, vulnerability analysis, and network security.
- Research operating system internals, kernel architecture, memory management, process scheduling, and low-level security mechanisms.
- Design and experiment with security monitoring infrastructure, intrusion detection systems, and network traffic analysis solutions.
- Maintain independent technical projects and publish research findings through personal development and malware analysis platforms.

MALWARE ANALYSIS & REVERSE ENGINEERING

- Perform static and dynamic analysis of Windows PE and Linux ELF binaries to investigate malicious behaviour and identify potential threats.
- Conduct binary analysis, disassembly, debugging, and reverse engineering to understand executable behaviour.
- Investigate malware persistence, process behaviour, API interactions, filesystem modifications, and network communication.
- Analyse indicators of compromise (IOCs), suspicious execution patterns, and malware techniques.
- Publish independent malware research, technical investigations, and reverse engineering write-ups.

Research Blog: r4r00t-blog.vercel.app

EXPLOIT DEVELOPMENT & VULNERABILITY RESEARCH

- Research software vulnerabilities, memory corruption, and low-level exploitation techniques.
- Investigate stack-based and heap-based vulnerabilities, buffer overflows, and return-oriented programming.
- Develop proof-of-concept exploits in controlled research environments.
- Examine binary protections, memory layouts, exploitation mitigations, and system-level vulnerabilities.

SELECTED TECHNICAL PROJECTS

NanoSec OS – Independent Operating System Development *Ongoing*

Independently developing a custom operating system from scratch using **C and x86 Assembly**, featuring a custom kernel and Unix-like computing environment.

- Implemented kernel components including preemptive multitasking, round-robin scheduling, virtual memory management, paging, and system calls.
- Developed a custom Unix-like shell with 50+ commands and an integrated scripting language, Nash.
- Implemented networking components covering TCP/IP, ARP, ICMP, UDP, TCP, and DNS.
- Developed security mechanisms including user authentication, packet-filtering firewall, ASLR, stack canaries, and audit logging.
- Worked on filesystem support, hardware drivers, graphics infrastructure, and desktop environment.

GitHub: NanoSec OS

Petti – Linux Container Runtime Development *Ongoing*

Developing a Docker-like container runtime from scratch using **Go**, implementing Linux containerisation primitives and OCI-compatible image management.

- Implemented Linux namespace isolation, process management, and resource control using cgroups v2.
- Developed layered filesystem support using OverlayFS and OCI image pulling with content-addressed storage.
- Implemented rootless container execution using user namespaces and container networking using virtual Ethernet interfaces.
- Integrated security mechanisms including capability dropping, seccomp filtering, and filesystem isolation through pivot_root.
- Implemented container lifecycle management, image build and save/load functionality, and Compose-lite orchestration.

GitHub: Petti

D4DoS – Distributed Denial-of-Service Detection System *Ongoing*

Developing a network security and DDoS detection platform using **Python and Django**.

- Developing network traffic inspection and anomaly detection capabilities.
- Integrating machine-learning-based detection approaches to identify abnormal traffic patterns.
- Designing a monitoring platform with cross-platform support and security event visibility.

WayRecon – Automated OSINT & Reconnaissance Tool *Sep – Oct 2025*

Developed a Python-based reconnaissance utility integrating the Wayback Machine CDX API to retrieve and analyse historical URLs.

- Implemented URL filtering, data extraction, export functionality, and structured JSON output.
- Designed a terminal-based workflow for automated reconnaissance and historical asset discovery.

SECURITY INFRASTRUCTURE & SYSTEM ADMINISTRATION

- Configured security monitoring infrastructure using **Wazuh, Suricata IDS, Elasticsearch, Prometheus, and Grafana**.
- Managed Linux-based VPS and dedicated server environments, including deployment, service management, and infrastructure administration.
- Configured web server environments using **Nginx and Apache**, alongside Docker-based deployments and Plesk administration.
- Worked with network monitoring, intrusion detection, firewall configuration, log analysis, and server security.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, C#, Java, Kotlin, Swift, Go, Rust, Ruby, Dart, R, Julia, Perl, Lua, Objective-C, JavaScript, TypeScript, PHP, Solidity, Elixir, Scala

Systems Programming & Assembly: x86 Assembly, x86-64 Assembly, ARM Assembly, Kernel Development, Systems Programming, Memory Management, Process Scheduling

Scripting: Bash, Zsh, PowerShell, Batch

Web Technologies: HTML5, CSS3, REST APIs, GraphQL, WebSockets

Cybersecurity: Penetration Testing, Vulnerability Research, Exploit Development, Malware Analysis, Reverse Engineering, Threat Hunting, OSINT, Digital Forensics, Network Security, Red Teaming, Blue Teaming, Incident Response

Malware Analysis: Ghidra, IDA Pro, x64dbg, WinDbg, GDB, Radare2, YARA, capa, PE-bear, Detect It Easy, FLOSS, Procmon, Process Explorer, API Monitor, Static Analysis, Dynamic Analysis, Binary Analysis

Exploit Development: pwntools, GDB, pwndbg, GEF, ROPgadget, Ropper, checksec, AFL++, libFuzzer, AddressSanitizer, Buffer Overflow Analysis, ROP, Memory Corruption

Offensive Security: Metasploit, Burp Suite, OWASP ZAP, Nmap, Masscan, RustScan, Amass, Subfinder, Assetfinder, FFUF, Gobuster, Feroxbuster, SQLmap, Nuclei, Nikto, Impacket, BloodHound, NetExec, Responder, Hashcat, John the Ripper, Hydra, Aircrack-ng, Bettercap, Shodan, Censys

Defensive Security: Wazuh, Suricata, Snort, Zeek, Wireshark, Splunk, Elastic Security, ELK Stack, Security Onion, YARA, Volatility, Velociraptor, osquery, Sysmon, MISP, OpenCTI, MITRE ATT&CK, VirusTotal

Infrastructure & DevOps: Docker, Docker Compose, Kubernetes, Podman, VMware, VirtualBox, KVM, QEMU, Proxmox, Terraform, Ansible, Jenkins, GitHub Actions, GitLab CI/CD, AWS, Azure, Google Cloud, Cloudflare, Hetzner

Networking: TCP/IP, DNS, DHCP, HTTP/HTTPS, SSH, VPN, WireGuard, OpenVPN, Tailscale, ZeroTier, Firewall Configuration, Network Traffic Analysis

Web Servers & Hosting: Nginx, Apache, LiteSpeed, Plesk, cPanel, DirectAdmin

Databases: MySQL, PostgreSQL, MongoDB, SQLite, MariaDB, Redis, Cassandra, Elasticsearch, OpenSearch, InfluxDB, Firebase, Supabase, Memcached

Messaging & Streaming: Apache Kafka, RabbitMQ, ActiveMQ, NATS, MQTT, Apache Pulsar, Kafka Streams

Data Engineering: Apache Spark, Apache Flink, Apache Hadoop

Monitoring & Observability: Prometheus, Grafana, Zabbix, Nagios, Netdata, Loki, OpenTelemetry

Frameworks & Libraries: Django, Flask, FastAPI, Django REST Framework, React, Next.js, Node.js, Express.js, NestJS, Vue.js, Angular, Svelte, Electron.js, Flutter, TensorFlow, PyTorch, OpenCV, NumPy, Pandas, Scikit-learn

Development Tools: Git, GitHub, GitLab, VS Code, GCC, Clang, LLVM, NASM, CMake, Make, GDB, QEMU

CERTIFICATIONS & PROFESSIONAL TRAINING

CAPT – Certified Associate Penetration Tester *Hackviser, Sep – Oct 2025*

Advanced Diploma in Cyber Defence *RedTeam Hackers Academy, May 2025 – Aug 2026*

OCSA – Offenso Certified Security Analyst *Offenso Hackers Academy, Jan – Apr 2025*

EDUCATION

Santhigiri Vidyabhavan Higher Secondary School *2025*

Senior Secondary (XII), Kerala Board of Higher Secondary Education – Arts

Academic Score: 89%

ACHIEVEMENTS

- Ranked **#1 in the UK region and #12 globally** on TryHackMe's monthly cybersecurity leaderboard.
- Secured **3rd place** in the Information Security Research Association (ISRA) offline CTF competition.

LANGUAGES

English (Professional) Malayalam (Native) Hindi (Intermediate)